MIL-H-35077B
23 September 1977
SUPERSEDING
MIL-H-35077A
17 March 1966

MILITARY SPECIFICATION

HORSERADISH, DEHYDRATED, GROUND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for dehydrated ground horseradish for use by the Department of Defense as an item of general issue.

1.2 Classification. The product shall be of the following classes, as specified (see 6.1):

Class 1 - See 3.2.1.1

Class 2 - See 3.2.1.2

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8950

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PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-G-460 - Glass Containers, Filled and Closed, Packaging and Packing of.

MILITARY

MIL-L-35078 - Loads; Unit: Preparation of Nonperishable Subsistence.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, PO Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The product shall be prepared from firm, sound, first quality horseradish roots which are practically free from decay spots, freezing damage, insect injury or other physical damage.

3.2 Finished product. The dehydrated product shall be practically free from bitterness, and it shall be free from scorched, musty or other objectionable odors. The finished product shall be free from adulterants such as turnips, parsnips, etc. When 250 milliliters (ml) of water are added to 5 grams of the powder, the resultant mixture shall yield a strong volatile pungency characteristic of fresh horseradish and shall contain not less than 0.9 percent allyl isothiocyanate. The color of the dehydrated product shall be between No. 37855 and No. 37886 of FED-STD-595. The product shall contain not more than 5.0 percent moisture and not more than 0.5 percent ash insoluble in acid.

3.2.1 Screen size.

3.2.1.1 Class 1. Class 1 shall be of such size that at least 98 percent, by weight, of the product shall pass through a US Standard No. 50 mesh screen.

3.2.1.2 Class 2. Class 2 shall be of such size that at least 95 percent, by weight, of the product shall pass through a US Standard No. 20 mesh screen, with not more than 25 percent, by weight, passing through a US Standard No. 40 mesh screen, and not more than 5 percent, by weight, passing through a US Standard No. 60 mesh screen.

* 3.3 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

3.4 Workmanship. The product shall be processed in establishments meeting the requirements of MIL-STD-668. The product shall be prepared and processed with minimum delay between the various stages of production.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own, or any other facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant sanitation inspection. Products shall be rejected if produced in plants which fail to meet sanitary requirements of 3.4.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Testing of components. Testing criteria for caps shall be as shown in table I. Test failure shall indicate an unacceptable lot of caps and use of such shall be cause for rejection of the involved end item.

TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and reqmt. para.
Caps	One cap	Caps	S-1	Tin plate	Nearest 0.01 lb/base box	4.4.2 and 5.1

4.3.2 In-process examination. Examination shall be performed during processing to determine if the horseradish is being properly trimmed and screened to remove pathologic and physical damage. In addition, examination shall be performed to determine if additional ingredients are being incorporated into the product. Use of damaged horseradish or the adding of ingredients to the product shall be cause for rejection of the involved lot.

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4.3.3 Examination of end item. The end item shall be examined in accordance with tables III through V and 4.3.3.1 and 4.3.3.2. The sampling plans and acceptance criteria shall be as specified in table II. The acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units.

* TABLE II. Sampling plans and AQLs for the tables

Table	Inspection level	Sample unit	Expression of lot size	AQLs Major A	Major B	Minor
III	I	One filled and closed primary container	Primary containers	0.25	1.5	4.0
IV	S-3	One filled and closed primary container	Primary containers	-	-	2.5
V	S-2	One filled and closed primary container	Primary containers	-	2.5	4.0

* TABLE III. Examination of primary container

Category			Defect
Major A	Major B	Minor	
101			Broken or cracked.
102			Cap or closure not as specified.
	151		Closure loose. 1/
	152		Chipped or thin spot.
	153		Coating of exterior or interior of cap missing, incomplete or can be abraded with fingernail.
	154		Liner of cap missing.
	155		Nomenclature missing, incorrect or illegible.
		201	Not clean.
		202	Blistered, cracked, peeled or loosened coating (pertains to cap or closure).
		203	Labeling information, other than nomenclature, missing, incorrect or illegible.

1/ For determination tightness of screw cap, place the bottle upright on top of a filled and closed shipping container, grasp the cap firmly with fingertips and turn cap gently counterclockwise with a minimum of downward pressure for one complete revolution. If the cap turns and the bottle remains stationary, a defect shall be scored.

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TABLE IV. Examination for net weight 1/ 2/

Category	Defect
Minor	
201	Net weight less than 2-7/16 ounces (69 grams).

1/ Report results to the nearest 1/16 ounces (1 gram).

2/ The lot shall be rejected if the sample data indicate a lot average net weight less than specified net weight.

TABLE V. Examination of product 1/

Category		Defect
Major B	Minor	Dehydrated:
	201	Color not within range specified.
151		Scorched, musty or other objectionable odor.
	202	Particle size not as specified. 2/
		Rehydrated: 3/
152		Bitter taste.
153		Does not have a strong volatile pungency characteristic of fresh horseradish.

1/ Presence of foreign material (e.g., metal, glass, stones, insect, insect parts, etc.) shall be cause for rejection of the lot.

2/ Procedure shall be in accordance with 4.4.3 and 4.4.3.1.

3/ Rehydrate the product in accordance with 3.2. Make one determination per sample unit.

* 4.3.3.1 Examination of packing, marking and shipping containers. Examination shall be in accordance with the criteria of PPP-G-460.

* 4.3.3.2 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078 and the applicable specification sheet for the type and class of load specified (see 6.1).

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4.3.4 Sampling procedure and acceptance criteria for testing of the end item. The finished product shall be tested for moisture, ash insoluble in acid, and allyl isothiocyanate in accordance with 4.4.1. The sample for testing shall be an 8-ounce (227 g) composite, derived from the number of primary containers indicated by inspection level S-2. Results shall be reported to the nearest 0.1 percent. Any test failure shall be cause for rejection of the lot.

4.4 Tests. Test procedures and controls which differ from those specified herein, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

* 4.4.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Spices and Other Condiments; Section: Spices:

Test	Method
Ash insoluble in acid	Ash insoluble in acid
Moisture	Moisture
Allyl isothiocyanate	Volatile oil in mustard seed (titration method)

4.4.2 Tin coating weight. Tin coating weight shall be tested in accordance with any method specified in PPP-C-96.

4.4.3 Sieving test. US Standard woven-wire-cloth sieves, which are 8 inches (203 mm) in diameter, have full-height frames and comply with RR-S-366, shall be used in making the sieving test. The sieves shall be fitted together in the following order:

Class 1:
Cover
US Standard No. 50 sieve
Pan

Class 2:
Cover
US Standard Nos. 20, 40 and 60 sieves
Pan

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4.4.3.1 Procedure. Weigh contents of container and place on the top sieve of an assembled set of specified sieves, including cover and catch pan. The left hand will hold the sieve assembly at an angle of about 30 degrees from the horizontal so that the upper edge of the tilted plane is toward the right hand. Shake by striking the right side of the sieve assembly with a short, sharp stroke with the right hand moving in a horizontal plane. Strike at the rate of about 150 strokes per minute. After each series of 25 strokes, turn the sieve assembly 1/6 of a revolution in the same direction. Continue shaking for 2 minutes, remove, weigh and calculate the percentages of the sample segregated by the sieving action.

5. PACKAGING

* 5.1 Packaging, level A. Two and one-half ounces (71 g) of the product shall be filled into a commercial glass jar. The jar shall be closed with a knurled screw cap or hermetic friction-type side seal closure cap with pry-off feature, made from not less than commercial 0.25-pound per base box electrolytic tin plate (see 4.4.2) and coated outside with a corrosion-resisting lacquer or enamel. The cap shall be coated inside with a suitable lacquer and contain a liner pad having a vinyl facing or any other combination of materials normally used for the product being packed. Knurled screw caps shall be secured in place with a shrink-on cellulose band or a strip of pressure-sensitive tape which overlaps itself.

* 5.2 Packing. Twenty-four jars of the product arranged 4 by 6 shall be packed on end in a snug-fitting intermediate container. The intermediate container shall be provided with liner, pads and partitions specified for four ounce (113 g) glass containers in accordance with PPP-G-460. Four intermediate containers shall be packed into a snug-fitting shipping container in accordance with level A or B as specified (see 6.1).

* 5.2.1 Level A. The intermediate containers shall be constructed and closed in accordance with style RSC, W5c of PPP-B-636. The shipping container shall be constructed, closed and reinforced in accordance with style RSC, V2s of PPP-B-636.

* 5.2.2 Level B. The intermediate container shall be constructed in accordance with 5.2.1 and shipping containers shall be constructed, closed and reinforced in accordance with style RSC, V3c, V3s or V4s of PPP-B-636.

* 5.2.2.1 Nonmetallic reinforcement of intermediate and shipping containers. When specified (see 6.1 and 6.3), containers specified in 5.2.1 and 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636 except that for V3c, V3s or V4s boxes, two reinforcing bands may be used, one lengthwise and one girthwise.

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* 5.2.3 Unit loads. When specified (see 6.1), the product, packed as specified above, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.3 Marking.

5.3.1 Glass containers. Commercial labeling, complying with Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, will be acceptable. In addition, a recipe for reconstitution shall be included on the label (see 3.2).

* 5.3.2 Intermediate containers and shipping containers. Intermediate containers and shipping containers shall be marked in accordance with MIL-STD-129 and PPP-G-460.

* 5.3.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of product required (see 1.2).
(c) Level of packing required (see 5.2).
(d) When packing specified in 5.2.2.1 is required.
(e) Type and class of unit load when unit loading is specified (see 5.2.3).

* 6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipping, handling, or storage of the specified item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUP PUB 740/AFR 71-6/ MCO 4030, 14D/DSAR 4145.7.

* 6.3 Intended use of nonmetallic reinforcement of shipping container. Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific overseas operations.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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